

CHAPTER 4

Propagation of states and covariances

In this chapter, we will begin with our mathematical description of a dynamic system, and then derive the equations that govern the propagation of the state mean and covariance. The material presented in this chapter is fundamental to the state estimation algorithm (the Kalman filter) that we will derive in Chapter 5.

Section 4.1 covers discrete-time systems. Section 4.2 covers sampled-data systems, which are the most common types of systems found in the real world. In this type of system, the system dynamics are described by continuous-time differential equations, but the control and measurement signals are discrete time (e.g., control based on a digital computer and measurements obtained at discrete times). Section 4.3 covers continuous-time systems.

4.1 DISCRETE-TIME SYSTEMS

Suppose we have the following linear discrete-time system:

$$x_k = F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \quad (4.1)$$

where u_k is a known input and w_k is Gaussian zero-mean white noise with covariance Q_k . How does the mean of the state x_k change with time? If we take the expected value of both sides of Equation (4.1) we obtain

$$\begin{aligned} \bar{x}_k &= E(x_k) \\ &= F_{k-1}\bar{x}_{k-1} + G_{k-1}u_{k-1} \end{aligned} \quad (4.2)$$

How does the covariance of x_k change with time? We can use Equations (4.1) and (4.2) to obtain

$$\begin{aligned}
 (x_k - \bar{x}_k)(\cdots)^T &= (F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} - \bar{x}_k)(\cdots)^T \\
 &= [F_{k-1}(x_{k-1} - \bar{x}_{k-1}) + w_{k-1}][\cdots]^T \\
 &= F_{k-1}(x_{k-1} - \bar{x}_{k-1})(x_{k-1} - \bar{x}_{k-1})^T F_{k-1}^T + w_{k-1}w_{k-1}^T + \\
 &\quad F_{k-1}(x_{k-1} - \bar{x}_{k-1})w_{k-1}^T + w_{k-1}(x_{k-1} - \bar{x}_{k-1})^T F_{k-1}^T
 \end{aligned} \tag{4.3}$$

We therefore obtain the covariance of x_k as the expected value of the above expression. Since $(x_{k-1} - \bar{x}_{k-1})$ is uncorrelated with w_{k-1} , we obtain

$$\begin{aligned}
 P_k &= E[(x_k - \bar{x}_k)(\cdots)^T] \\
 &= F_{k-1}P_{k-1}F_{k-1}^T + Q_{k-1}
 \end{aligned} \tag{4.4}$$

This is called a discrete-time Lyapunov equation, or a Stein equation [Ste52]. We will see in the next chapter that Equations (4.2) and (4.4) are fundamental in the derivation of the Kalman filter.

It is interesting to consider the conditions under which the discrete-time Lyapunov equation has a steady-state solution. That is, suppose that $F_k = F$ is a constant, and $Q_k = Q$ is a constant. Then we have the following theorem, whose proof can be found in [Kai00, Appendix D].

Theorem 21 *Consider the equation $P = FPF^T + Q$ where F and Q are real matrices. Denote by $\lambda_i(F)$ the eigenvalues of the F matrix.*

1. *A unique solution P exists if and only if $\lambda_i(F)\lambda_j(F) \neq 1$ for all i, j . This unique solution is symmetric.*
2. *Note that the above condition includes the case of stable F , because if F is stable then all of its eigenvalues are less than one in magnitude, so $\lambda_i(F)\lambda_j(F) \neq 1$ for all i, j . Therefore, we see that if F is stable then the discrete-time Lyapunov equation has a solution P that is unique and symmetric. In this case, the solution can be written as*

$$P = \sum_{i=0}^{\infty} F^i Q (F^T)^i \tag{4.5}$$

3. *If F is stable and Q is positive (semi)definite, then the unique solution P is symmetric and positive (semi)definite.*
4. *If F is stable, Q is positive semidefinite, and $(F, Q^{1/2})$ is controllable, then P is unique, symmetric, and positive definite. Note that $Q^{1/2}$, the square root of Q , is defined here as any matrix such that $Q^{1/2}(Q^{1/2})^T = Q$.*

Now let us look at the solution of the linear system of Equation (4.1):

$$x_k = F_{k,0}x_0 + \sum_{i=0}^{k-1} (F_{k,i+1}w_i + F_{k,i+1}G_i u_i) \tag{4.6}$$

The matrix $F_{k,i}$ is the state transition matrix of the system and is defined as

$$F_{k,i} = \begin{cases} F_{k-1}F_{k-2}\cdots F_i & k > i \\ I & k = i \\ 0 & k < i \end{cases} \quad (4.7)$$

Notice from Equation (4.6) that x_k is a linear combination of x_0 , $\{w_i\}$, and $\{u_i\}$. If the input sequence $\{u_i\}$ is known, then it is a constant and can be considered to be a sequence of Gaussian random variables with zero covariance. If x_0 and $\{w_i\}$ are unknown but are Gaussian random variables, then x_k in Equation (4.6) is a linear combination of Gaussian random variables. Therefore, x_k is itself a Gaussian random variable (see Example 2.4). But we computed the mean and covariance of x_k in Equations (4.2) and (4.4). Therefore

$$x_k \sim N(\bar{x}_k, P_k) \quad (4.8)$$

This completely characterizes x_k in a statistical sense since a Gaussian random variable is completely characterized by its mean and covariance.

■ EXAMPLE 4.1

A linear system describing the population of a predator $x(1)$ and that of its prey $x(2)$ can be written as

$$\begin{aligned} x_{k+1}(1) &= x_k(1) - 0.8x_k(1) + 0.4x_k(2) + w_k(1) \\ x_{k+1}(2) &= x_k(2) - 0.4x_k(1) + u_k + w_k(2) \end{aligned} \quad (4.9)$$

In the first equation, we see that the predator population causes itself to decrease because of overcrowding, but the prey population causes the predator population to increase. In the second equation, we see that the prey population decreases due to the predator population and increases due to an external food supply u_k . The populations are also subject to random disturbances (with respective variances 1 and 2) due to environmental factors. This system can be written in state-space form as

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 0.2 & 0.4 \\ -0.4 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_k + w_k \\ w_k &\sim (0, Q) \quad Q = \text{diag}(1, 2) \end{aligned} \quad (4.10)$$

Equations (4.2) and (4.4) describe how the mean and covariance of the populations change with time. Figure 4.1 depicts the two means and the two diagonal elements of the covariance matrix for the first few time steps when $u_k = 1$ and the initial conditions are set as $\bar{x}_0 = [10 \ 20]^T$ and $P_0 = \text{diag}(40, 40)$. It is seen that the mean and covariance eventually reach steady-state values given by

$$\begin{aligned} \bar{x} &= (I - F)^{-1}Gu \\ &= [2.5 \ 5]^T \\ P &\approx \begin{bmatrix} 2.88 & 3.08 \\ 3.08 & 7.96 \end{bmatrix} \end{aligned} \quad (4.11)$$

The steady-state value of P can also be found directly (i.e., without simulation) using control system software.¹ Note that since F for this example is stable and Q is positive definite, Theorem 21 guarantees that P has a unique positive definite steady-state solution.

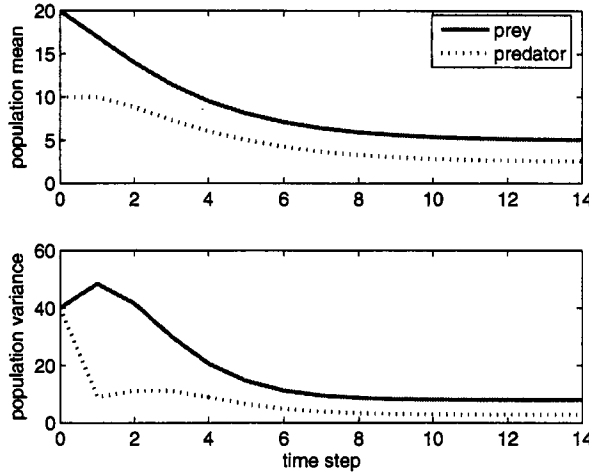


Figure 4.1 State means and variances for Example 4.1.

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In Equation (4.1), we showed the process noise directly entering the system dynamics. This is the convention that we use in this book. However, many times process noise is first multiplied by some matrix before it enters the system dynamics. That is,

$$x_k = F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + L_{k-1}\tilde{w}_{k-1}, \quad \tilde{w}_k \sim (0, \tilde{Q}_k) \quad (4.12)$$

How can we put this into the conventional form of Equation (4.1)? Notice that the rightmost term of Equation (4.12) has a covariance given by

$$\begin{aligned} E[(L_{k-1}\tilde{w}_{k-1})(L_{k-1}\tilde{w}_{k-1})^T] &= L_{k-1}E(\tilde{w}_{k-1}\tilde{w}_{k-1}^T)L_{k-1}^T \\ &= L_{k-1}\tilde{Q}_{k-1}L_{k-1}^T \end{aligned} \quad (4.13)$$

Therefore, Equation (4.12) is equivalent to the equation

$$x_k = F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1}, \quad w_k \sim (0, L_k\tilde{Q}_kL_k^T) \quad (4.14)$$

This idea is illustrated in Sections 7.3.1 and 7.3.2. The same type of transformation can be made with noisy measurement equations. That is, the measurement equation

$$y_k = H_kx_k + L_k\tilde{v}_k, \quad \tilde{v}_k \sim (0, \tilde{R}_k) \quad (4.15)$$

is equivalent to the measurement equation

$$y_k = H_kx_k + v_k, \quad v_k \sim (0, L_k\tilde{R}_kL_k^T) \quad (4.16)$$

¹For example, we can use the MATLAB Control System Toolbox function `DLYAP(F, Q)`.